PAH-v2 plaquette paths freeze in every declared local clock

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1. Purpose and scope

This note resolves the random-path sampling question left open by PAH-TIME-001. It does not supply that audit's missing operational clock law. Fix exactly the admitted finite open square $G_{h,N}$, all its elementary faces, stored edge orientations and source anchors. Put $v = |V|$, $e = |E|$, $p = |P| > 0$. These are combinatorial counts, not physical dimensions or volume. All parent parameters are fixed, including $0 < \epsilon \le 1$ and $\beta, \nu > 0$, $R_0 > 0$, and integers $1 \le q_0 \le m_0$. Only r varies:

$$K_r = \prod_{i=1}^{r+1} p_i, \quad M_s = 2^r, \quad M_\psi = m_0 4^r, \quad R_{\max} = R_0 2^r, \quad Q = q_0 2^r. \tag{1}$$

Every integer counting coordinate and signed PH/TR/LK/AP incidence is retained. In particular hidden labels at zero radius or $\epsilon = 1$, and the two $K = 2$ labels with equal endpoints, are not identified. The unchanged finite generator, full domain and initial state are

$$\begin{aligned} L_r f(x) &= \sum_{a:\, ax \text{ valid}} c_a(x)[f(ax) - f(x)], \quad c_a(x) = m_a(x) e^{-\beta(F_r(ax) - F_r(x))/2}, \\ D(L_r) &= \mathbb{C}^{\Omega_r}, \qquad X_0 \sim \pi_r = Z_r^{-1} e^{-\beta F_r}. \end{aligned} \tag{2}$$

X_t is the original right-continuous finite Markov path on $t \ge 0$, with no terminal conditioning. No altered dynamics is adopted. The readout and the already declared local path clocks, with zero origin, are

$$W_r(x) = p^{-1} \sum_{b \in P} \operatorname{Re} U_b(x), \qquad \tau_v(t) = \int_0^t s_v(X_u)^\nu \, du. \tag{3}$$

W is dimensionless, unweighted and gauge/anchor invariant. The clocks are gauge neutral and covariant with vertex relabelling. No preferred vertex or synchronized physical clock is selected. Write σ_v for the inverse of τ_v; its existence is proved below.

Preregistered target. For every fixed finite $T \ge 0$, prove

$$E_r(T) := \mathbb{E}_{\pi_r}\left[\max_{v \in V} \sup_{0 \le u \le T} |W_r(X_{\sigma_v(u)}) - W_r(X_0)|^2\right] \longrightarrow 0. \tag{4}$$

This is an original-path random sampling diagnostic, not substitution of a random time into $e^{tL_r}W$. No common state/path space across r, CRT repair or limiting process is asserted. The preregistration is

`strategy/pa-hyp/PAH-TIME-002-prereg-v1.json`
`a4d3700b323f4b52664cc01680749f4e359c0854bc96f9bf77527a50a827ba10`

2. Source cap and exact labelled path realization

Use only the fixed-scope full-state estimate of R-575, equations (5)–(7): $\sum_v |\psi_v| = A = R_0 q_0/m_0$, $|F_r(x)| \le B$, and

$$\begin{aligned} b_0 &= \lambda_s(1-\epsilon)^2/2 + |m^2|A^2/2 + |\lambda_4|A^4/4 + \eta_6 A^6/6 + gA^2/2, \\ B &= vb_0 + \kappa_s e(1-\epsilon)^2/2 + 2\kappa_D e A^2/\epsilon + 2\kappa_g p/\epsilon, \qquad C = e^{\beta B}. \end{aligned} \tag{5}$$

The parameter domains and normalization are identical. These bounds follow on every full tuple from $|\psi_w - U_e\psi_v| \le 2A$, $J_e, J_b \le 1/\epsilon$, and $0 \le 1 - \operatorname{Re} U_b \le 2$. Every mobility is at most one; hence $c_a(x) \le C$, including signed mass/quartic coefficients. C is independent of r, not of geometry, ϵ or the other parameters.

There are $M = 4(v+e)$ potential labels. Give each label an independent Poisson proposal stream of rate C, with independent uniform acceptance marks. At a proposal, apply its root with probability $c_a(x)/C$ if valid; otherwise stay. Rejected proposals are auxiliary bookkeeping, not new source events. Invalid roots have acceptance zero and are never evaluated outside Ω_r. The infinitesimal generator of the resulting state path is exactly (2). Equivalently its one-proposal kernel P satisfies

$$Pf(x) - f(x) = M^{-1} \sum_{a \text{ valid}} (c_a(x)/C)[f(ax) - f(x)], \qquad MC(P-I) = L_r. \tag{6}$$

The proposal sum is finite on compact intervals almost surely. Conditional on its count, the kernel is P, so the transition matrices are $e^{-MCt}\sum_{n\ge 0}(MCt)^n P^n/n! = e^{tL_r}$. Independent future proposal increments give the finite Markov path law with the stipulated initial distribution. Thus this construction represents the original chain; it does not change rates or time. Duplicate labels contribute their separate rates, as required by the source.

Let $N_{\mathrm{LK}}^{\mathrm{prop}}(S)$ count all LK proposals in $[0,S]$. It is Poisson with mean ΛS, where $\Lambda = 2eC$, because there are exactly $2e$ independent signed LK proposal streams. The accepted LK events need not be Poisson: their intensities depend on the full state.

3. Maximal path estimate and all local clocks

PH, TR and AP leave u_e, hence instantaneous W, unchanged. They still affect subsequent LK rates; (5)–(6) do not discard them. A single LK move touches at most two elementary faces, so telescoping their unit holonomies gives, with modular chord distance,

$$|\Delta_a W_r| \le \frac{2}{p}|e^{2\pi i/K_r} - 1| = \frac{4}{p}\sin(\pi/K_r) \le d_r := \frac{4\pi}{pK_r}. \tag{7}$$

This counts boundary edges safely and retains both signs at $K = 2$. Telescoping along each original path implies

$$\sup_{0\le t\le S} |W_r(X_t) - W_r(X_0)| \le d_r N_{\mathrm{LK}}^{\mathrm{prop}}(S). \tag{8}$$

For $N \sim \mathrm{Poisson}(z)$ the generating function is $\mathbb{E}q^N = e^{z(q-1)}$. Differentiating at $q = 1$ yields $\mathbb{E}N = z$, $\mathbb{E}N(N-1) = z^2$, and $\mathbb{E}N^2 = z + z^2$. Thus (8) has a square-integrable envelope.

Since $\epsilon^\nu \le s_v(X_t)^\nu \le 1$, each clock in (3) is continuous, strictly increasing and onto $[0,\infty)$, with

$$\epsilon^\nu t \le \tau_v(t) \le t, \qquad u \le \sigma_v(u) \le u/\epsilon^\nu. \tag{9}$$

The inverse is causal: $\{\sigma_v(u) \le t\} = \{\tau_v(t) \ge u\}$. No stochastic time-change or stationarity theorem is required. For $S = T/\epsilon^\nu$ every sampled time in (4), for every vertex, lies in the same $[0,S]$. Consequently, without a vertex union factor,

$$E_r(T) \le d_r^2\{\Lambda S + (\Lambda S)^2\} = \left(\frac{4\pi}{pK_r}\right)^2 \left\{2eCS + (2eCS)^2\right\} \longrightarrow 0. \tag{10}$$

All paths are piecewise constant with finitely many proposals on $[0, S]$; the finite-vertex maxima and time suprema above are measurable. They are bounded by (8) and also by 2 in absolute displacement. The limit in (10) uses $K_r \geq 2^{r+1} \to \infty$ and fixed C, e, p, S. It is a second-moment bound of order K_r^{-2}, not a derived relaxation time or its lower bound.

4. Independent checks of the analytic step

An independent derivation avoids Poisson independence. Count accepted labelled LK incidences by N_t; their predictable intensity obeys $\lambda_r(X_t) \leq \Lambda$. Stop at the mth LK incidence and freeze the augmented finite state/count chain. Its generator on n and n^2 is, respectively, $1_{n<m}\lambda_r(x)$ and $1_{n<m}(2n+1)\lambda_r(x)$. Integrating the finite matrix exponential gives

$$\mathbb{E}N_t^{(m)} \leq \Lambda t, \qquad \mathbb{E}(N_t^{(m)})^2 \leq \int_0^t \Lambda(2\mathbb{E}N_s^{(m)} + 1)\, ds \leq \Lambda t + \Lambda^2 t^2. \tag{11}$$

The full rate cap MC gives nonexplosion. Monotone convergence as $m \to \infty$ and pathwise telescoping give (10) again. No new physical carrier is introduced: the finite stopped count is a proof device only.

A second, weaker check uses original external-time stationarity and the per-label midpoint flux bound from R-575 equation (11): $\sum_x \pi_r(x)c_a(x) \leq 1$. For LK it follows by Cauchy–Schwarz on the bijective labelled root from $\pi_r c_a = m_a\sqrt{\pi_r(x)\pi_r(ax)}$. Expected total variation on $[0, S]$ is at most $2ed_rS$. Because $W \in [-1, 1]$, the squared maximum displacement is at most twice that variation. Hence

$$E_r(T) \leq 4ed_rS \longrightarrow 0. \tag{12}$$

Stationarity is used here only at deterministic original times before the pathwise interval containment, never in the sampled clock. The positive initial Gibbs variance of W in R-575 remains separate; (10) is not a constant-readout artifact, nor a new stationary local-clock law.

5. Applicability and verification boundary

Ward Whitt, *Continuous-Time Markov Chains*, December 19, 2006, section 3.4, equations (3.24)–(3.28) and Theorem 3.4, pp. 14–15, is the primary reference for finite uniformization. The source URL and access date are in the preregistration. Finite state, nonnegative rates, bounded total rate and identical initial distribution hold here; (6) derives the labelled specialization. No infinite-state or ergodic conclusion is imported.

R-570's finite full-domain result and R-575's cap apply to identical functional/state/root/schedule bytes. R-575's deterministic semigroup bound alone does not prove (4); (8)–(9) are the new bridge. The old R-573/R-574 CRT-cylinder counterexamples remain unchanged. TIME001's operational-clock identification remains missing.

Primary exact source/symbolic controls, non-importing independent incidence/count controls, and 16 hostile algebra/metadata controls have separate run JSONs. The width-four, $K = 6$ hostile readout fixture uses the admitted schedule; no new dynamics table or carrier is studied. The bound for arbitrary r is proved above, not inferred from fixtures. Five parameterized Lean declarations check moment algebra, squared domination, inverse horizon, an integrable expectation inequality and limit domination. The actual CTMC, Poisson moments, source estimates, clock inverse construction and K_r divergence are not fully encoded. Reviewers are separate in-session agents, not external-person referees. External adversarial replay is invited.

6. Devil's-advocate review

1. **Only one sign counted. DISMISSED:** $2e$ LK labels are retained, including duplicate $K = 2$ endpoints. There is no rate division by multiplicity.

2. **One affected face assumed. DISMISSED:** the factor two in (7) is necessary. On admitted width four, $K = 6$, a shared-link fixture has displacement $2/9$ and chord one; the false bound $1/9$ fails.

3. **Poisson variance omitted. DISMISSED:** the moment is $z + z^2$, not z^2. Accepted events are not assumed to be Poisson.

4. **Other roots were removed. DISMISSED:** every signed root is present in (6); they may change future LK rates. Only their instantaneous W increments vanish.

5. **Random time inserted into a semigroup. DISMISSED:** (8) controls entire original paths; (9) is deterministic interval containment on each path.

6. **A new Gibbs state is needed. UPHELD for a changed-clock stationarity claim:** none is made. The initial law stays π_r; (10) actually requires no stationary initial law.

7. **An arbitrary vertex defines a physical clock. DISMISSED:** (4) ranges over every vertex covariantly; it chooses no global observation clock.

8. **Small aperture or infinite volume evades the constants. UPHELD as a scope limit:** S and B need not be uniform in those limits. Neither a joint limit nor a kinetic acceleration is authorized or tested.

9. **Source or physical meaning repaired. UPHELD as a prohibition:** no CRT repair, common limit, operational calibration or physical identity follows. This is a model-specific obstruction using standard finite tools.

7. Result footer

```
Result ID: R-576; PAH-TIME-002; exploration EXP-001834.
Precise statement: Equation (10) proves the preregistered maximal
original-path W freezing simultaneously in every declared local clock.
Scope: Fixed admitted finite square and parameters, epsilon>0;
original full counting Gibbs initial law and external-time path;
only the approved local-state cutoff r tends to infinity.
Dependencies: Frozen parent functional/rates, v2-r2 labelled roots,
adopted schedule, R-575 cap; finite labelled probability construction.
Evidence grade: ANALYTIC, EXACT, EXECUTED, INHERITED cap.
Reproduction command: python -X utf8
  verification/scripts/pah_time002_verify.py
  --lean-cache E:/Dev/TECT/verification/lean/.lake/packages
  --elan-home C:/Users/NaEun/.elan
Expected output: primary, independent, hostile, partial Lean,
and integrated PASS. Assemble the command above on one line.
Falsification gate: exact source/cap/root/path/clock or moment error.
Tier before / after: C6 T1 OPEN -> T1 OPEN; T-054 gate unchanged.
No-overclaim statement: no physical Pre-A, spacetime, QFT, gravity,
horizon, continuum, mass gap or TOE; no operational-clock admission.
Next required action: finish checkpoint, then park. Reopen only for
an exact error or a separately admitted, independently motivated
operational clock/readout packet outside this fixed-scope obstruction.
No automatic proof chain, rate fitting, external enquiry or rescaling.
```